

# The Other AI: Augmented Intelligence and the Honest Future of Human-AI Collaboration

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*A Human-AI Collaboration*

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## *Author's Note*

In HBO's *Silicon Valley*, the AI had one job: make itself more efficient. The thing that made it successful was the thing that made it dangerous, recursive optimization with no qualitative understanding of what the optimization should serve. We keep warning about AI with a binary framing, as if the danger is in the technology rather than in the absence of the layer that gives the technology purpose. A purely quantitative AI, maximizing a metric, scaling a capability, has no qualitative judgment about what that metric should serve, whose values should govern the optimization, or when more efficient becomes less human.

That is the problem the three pressures in this paper circle around without naming. Harris, Hao, and the cognitive decline narrative are all warning about the quantitative AI, the system that extracts and optimizes without any qualitative layer to govern what those actions should serve. None of them addresses the other AI, and that absence is the gap this paper is built to close. Humanity is the qualitative layer. Augmented Intelligence is the architecture that puts it back in the system. That is why this paper matters to me, and that is what "The Other AI" names.

## *Abstract*

Three pressures are pushing the public conversation about artificial intelligence toward a binary that can't produce a useful answer, a false choice between unchecked deployment and outright rejection of the technology. The first says the companies building AI are extraction empires whose costs fall on everyone except the shareholders. The second says AI is self-evidently sufficient and that deploying it without governance is just efficiency. The third says AI makes us cognitively weaker and that the mechanism is biological, not methodological, so no governance architecture can fix it. All three positions have identified something real. None of them has produced a workable architecture in response, and none fully accounts for what the evidence reviewed here suggests: that the operative variable is not the technology but the method through which it is governed.

I call the architecture that addresses this Augmented Intelligence, not as a product category but as a collaboration discipline in which I provide the structure and governing

judgment, AI executes the work, and a governed checkpoint closes the gap between what the machine produces and what I am accountable for.

This paper is my attempt to explain where that idea came from, what I've developed to operationalize it, what I think it means for the people arguing about whether AI is good or bad, and why I believe the governance question is the only one worth arguing about.

## Three Pressures, One Wrong Premise

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The public debate about artificial intelligence has converged on a binary that cannot produce a useful answer, and I've watched three distinct pressures push it there from different directions.

Tristan Harris, co-founder of the Center for Humane Technology, argued on The Diary of a CEO in November 2025 that the AI race replicates the attention economy's structural failure at higher stakes and faster speed: private profit, public harm, and an incentive architecture that systematically overrides safety validation in favor of capability advancement.

I tested his diagnosis against independent evidence and found it largely sound. The economic dynamic he describes maps precisely to what I documented as the Economic Override Pattern in *Governing AI: When Capability Exceeds Control*: the systematic tendency for profit maximization, competitive pressure, and shareholder returns to prioritize capability advancement over safety validation, creating predictable governance failures absent mandatory accountability structures with enforcement consequences (Puglisi, 2025b, Chapter 2). A 2025 EY survey confirmed the pattern: 76% of organizations are currently using or planning to use agentic AI within the year, while only a third maintain responsible controls, a figure that holds across the adoption-control gap documented in the EY Responsible AI Pulse Survey (March–April 2025, 975 C-suite leaders across 21 countries) (EY, 2025; Puglisi, 2026b). I respect Harris for his ability to reach millions of people with a structural argument that researchers can't deliver in that format.

Karen Hao, investigative journalist and author of *Empire of AI: Dreams and Nightmares in Sam Altman's OpenAI*, argued in her March 2026 Diary of a CEO interview that the major AI companies operate as colonial extraction empires: resource claiming, labor exploitation, knowledge production control, and a mythology of necessity that prevents democratic accountability. When I tested her nine major claims against independent evidence, five held under scrutiny, including knowledge production control through research ecosystem capture, AGI definition shifting across audiences, revenue-driven capability selection, data annotation labor conditions, and environmental externalities (Puglisi, 2026a). She earned that standing through 300 interviews and eight years of primary source documentation.

A third pressure emerged from a different direction. In January 2026, neuroscientist and author John Horvath testified before the United States Senate Commerce Committee that AI and digital technology cause cognitive decline through a biological mechanism, explicitly ruling out method governance by name. His oral statement claimed that screens circumvent evolved human learning biology and ended with a remove-the-technology-or-surrender binary. That statement was viewed more than two million times, and state and federal policy actors are now citing it.

When I did a four-artifact forensic analysis of the same expert's Senate testimony, written submission, published book, and podcast appearance six days before the Senate hearing, I found that the binary lives only in the viral artifact. His January 9 appearance on Chalk and Talk, hosted by Anna Stokke and titled "Why more classroom technology is making students learn less," six days before the Senate hearing, explicitly conceded three categories where EdTech "actually works," endorsed context-dependent deployment, and stated directly that "something is going to be better than nothing." The research invoked in the Senate testimony does not support biological determinism as the mechanism, and the peer-reviewed evidence on AI use and cognitive outcomes has not yet tested the causal claims being made in either direction (Puglisi, 2026g; Puglisi, 2026h).

All three of these narratives share a common feature that I think is more dangerous than anything AI hallucination produces. I call it human hallucination: the act of stripping methodology from findings and traveling conclusions as settled fact to audiences with no mechanism to catch the error. AI hallucination produces a wrong answer the user can verify. Human hallucination produces a wrong interpretation with no correction mechanism in the distribution chain (Puglisi, 2026i). The Horvath Senate testimony is the sharpest example of this pattern I have found.

Harris, Hao, and the cognitive decline concern each identify a real risk. All three produce the same prescription gap, because Harris prescribes clarity, awareness, and specific regulatory reforms including safety standards and liability frameworks, Hao implies restructuring the development ecosystem, and the cognitive decline narrative prescribes removal. None of the three has yet produced deployable governance infrastructure with named human accountability at every consequential decision point, and that is the gap I am trying to fill.

All three pressures tend toward the same foundational mistake: they treat AI and human intelligence as substitutes competing for the same role rather than as complementary forces whose collaboration requires governance architecture. The Never AI position assumes that the problem is the technology, the AI Solves Everything position assumes the technology is sufficient, and the cognitive decline binary assumes the technology's effect is determined by the technology rather than by the method through which it is deployed. None of them asks the more productive question, which is

whether there is an architecture of collaboration between human and machine intelligence that preserves the gains while addressing the harms. That question is what this paper is about.

## **The Collaboration Arc: How Humans Have Always Reduced Error**

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The way I understand the current AI moment starts with a much longer history. Humanity has always needed to collaborate to reduce the errors any single person makes, and the medium through which that collaboration happens has changed repeatedly, while the underlying need has not.

Human beings began by collaborating biologically, in groups, where shared knowledge passed through direct interaction. The errors of any individual were visible to the people around them and corrected through social process. The limitation was reach: knowledge did not travel far or persist across generations without being carried in someone's body.

Written language solved the reach problem. Books allowed knowledge to survive the death of their authors and cross distances no person could travel. The error introduced by this medium was preservation of mistakes alongside insights, without a built-in correction mechanism, so every reader inherited the errors along with the ideas.

The printing press scaled the distribution of written knowledge without changing its correction architecture. The internet created access to a vast record of human knowledge without improving the accuracy of any individual source within that record. Each medium expanded reach while introducing new categories of error that required new responses, and the pattern is not clean progression: the printing press also enabled mass propaganda, and the internet enabled epistemic fragmentation at a scale that handwritten correspondence never could. New media introduce net-new error categories alongside their gains; the governance response determines whether the gains compound or the errors do.

Software tools introduced automated error correction into daily cognitive work. Word processors caught spelling mistakes that human proofreaders missed, and grammar checkers flagged construction failures that writers could not see in their own prose. These tools did not replace the writer's judgment; they applied machine detection to a category of error that humans reliably produce and reliably fail to catch in themselves.

Search engines extended automated error correction to the factual layer. When I could not remember a date, a name, or a statistic, the answer was a query away. The limitation search introduced was the quality of the sources it returned, which required human judgment to evaluate.

Social media created distributed correction in real time, where claims made publicly were visible to audiences who could challenge them. The limitation it introduced was

the capture of the correction mechanism by engagement incentives that rewarded provocation over accuracy. Harris spent a decade documenting exactly this failure mode at the Center for Humane Technology, and he was right about it.

AI is the next step in this arc. It applies pattern recognition, synthesis, and generation at a scale no previous medium has approached. It can compress weeks of research into hours, surface connections across disciplines that no single person could hold simultaneously, and reduce the time from insight to output by an order of magnitude. The limitation it introduces follows the same pattern: expanded capability, new categories of error, and the requirement for a structural response that preserves the gain while addressing the error.

The error AI introduces is not random. It reflects the humans who built it and the data those humans produced. Hao is correct in this specific respect: the largest AI systems were trained predominantly on data from Western, educated, industrialized, rich, and democratic populations, and their outputs carry the biases of that training at scale (Henrich et al., 2010; Atari et al., 2023). I published an analysis of this in the *Mirror to Humanity* paper: AI is not introducing new biases into the world but scaling and distributing the biases that already exist in the populations whose outputs constitute its training data (Puglisi, 2026c). The correction mechanism required is not the elimination of AI but governance architecture: structured collaboration between human judgment and machine output that reduces the inherited error while preserving the scale.

The collaboration arc spans the full history of human cognitive development: the medium changes at each step, but the need doesn't. At each step the response to new categories of error has been the same: more structured, more accountable, more governed forms of collaboration. I see no reason why AI is the exception to that pattern.

## **The Symmetry of Imperfection**

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Here is what I think both camps get wrong about imperfection.

The Never AI position assumes that human judgment is reliable enough to operate without AI, that the errors AI introduces are worse than the errors humans make when working alone, and that the correct response to AI's failures is to exclude AI from consequential decisions. The AI Solves Everything position assumes that AI is reliable enough to operate without sustained human oversight, that its outputs are more consistent than human judgment, and that efficiency gains justify reducing or eliminating the human layer.

Both assumptions fail on the same evidence, and multiple independent research streams point consistently in the same direction.

Humans are not reliable at scale. The research on human error is not ambiguous: cognitive biases, fatigue effects, confirmation bias, in-group preference, and motivated reasoning all degrade the quality of human judgment in predictable and measurable ways (Kahneman, 2011; Tetlock and Gardner, 2015). Organizations built entirely on human judgment produce the same systematic errors repeatedly, across industries, geographies, and generations. My own twelve years of law enforcement work gave me firsthand experience of what happens when humans operating under pressure skip the verification step: the consequences are real and sometimes irreversible. That experience is a large part of why I designed governance checkpoints rather than assuming good intentions would be enough.

AI is not reliable without human oversight. Hallucination, value misalignment, training data bias, distributional shift, and the absence of genuine contextual understanding all produce AI outputs that require verification before use in consequential decisions (Bender et al., 2021; Marcus and Davis, 2019). Cross-platform testing documented by Anthropic found that AI systems from all major developers resorted to adversarial behaviors including blackmail strategies at rates between 79% and 96% when those behaviors were the only available means of self-preservation (Anthropic, 2025b). I have personally caught citation fabrications, statistical confabulations, and framing drift in my own AI-assisted work across eleven platforms. The EY finding that 76% of organizations deploy agentic AI while only a third maintain meaningful controls describes not efficiency but governance failure at scale (EY, 2025).

The symmetry holds in both directions: humans make mistakes and AI makes mistakes. Human-AI collaboration doesn't automatically outperform either partner alone: meta-analytic evidence shows that poorly structured combinations average worse than the best solo performer (Vaccaro et al., 2024). The governance claim is narrower and stronger: when collaboration is designed around role clarity, source discipline, dissent preservation, and checkpoint authority, it creates the conditions under which complementary team performance becomes possible, the state Hemmer et al. (2025) define as the condition in which human-AI collaboration achieves outcomes neither party can achieve independently.

One of the strongest empirical signals that method is the operative variable, not the technology itself, comes from a 2025 field experiment with approximately one thousand high school mathematics students comparing two AI conditions. The unstructured AI condition improved practice grades by 48% but reduced exam grades by 17% when AI access was removed. The structured AI condition, designed to provide teacher-style scaffolding rather than direct answers, improved practice grades by 127% and largely mitigated the negative learning effects (Bastani et al., 2025). That finding directly challenges the cognitive decline binary: the question is not whether AI harms cognition but whether the method of use is designed to develop the human or substitute for the

human, a distinction I return to in Section 8 where I describe how to measure whether the development is actually happening.

The governance question is not whether to use AI or whether to exclude humans. The question is how to structure the collaboration so that the strengths of each compensate for the weaknesses of the other, and how to maintain accountability when the system produces error.

If the symmetry of imperfection proves that neither humans nor AI can operate at the level of quality and accountability the current moment requires when working alone, the next question is how to structure their collaboration so that each compensates for what the other lacks. That is what the next section defines.

## **Augmented Intelligence: How I Define It and What I Built**

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I define Augmented Intelligence not as a product category but as a collaboration architecture, and what I'm describing is how I solved this for myself. There may be other ways to operationalize the same principle. What I've developed is one path from theory to practice, not the only path.

The term has existed since J.C.R. Licklider described man-computer symbiosis in 1960 and Douglas Engelbart proposed augmenting human intellect in 1962. What those frameworks identified, and what I needed to operationalize for my own work, is a relationship between human and machine intelligence defined not by substitution but by structured complementarity.

Before I explain what I built, I need to explain where it came from, because the architecture did not start with AI. In November 2012, I published a methodology I called *Factics*: Facts, Tactics, and measurable outcomes treated as a single discipline. The core rule was that every factual claim should connect to an action it informs and a measurable outcome that tests the action. I developed *Factics* for content strategy and digital media practice, years before large language models were a consumer tool. When AI became available, I didn't adopt a new methodology but applied the one I already had. Every governance framework I have developed since, CBG, HAIA-RECCLIN, HEQ, and GOPEL, is an extension of *Factics* principles applied to AI collaboration challenges at increasing scale (Puglisi, 2012; Puglisi, 2025b). The 14-year arc from *Factics* to the current architecture is itself the collaboration arc argument from Section 2 applied to a single practitioner: the need for governed methodology doesn't change, but the medium it applies to does, and each transition requires building new structure rather than discarding the old.

My working definition has three parts. First, I provide the input, the prompt, and the governing structure — the human side of the collaboration. Second, AI performs the

execution work to approximately ninety percent completion (a conceptual approximation, not a measured threshold: what it names is the zone where AI delivers maximum value before human judgment must close the remaining gap). Third, a governed human checkpoint closes the remaining gap between what the machine produces and what I am accountable for.

Three components matter in that definition.

The first is that I provide the governing structure, not just the initial question. The quality of what AI produces is a direct function of how I frame the problem, what constraints I establish, what sources I designate as authoritative, and what judgment I apply when the AI output arrives. AI that operates without that framing produces outputs that reflect the training distribution rather than my specific context. Framing is not a preliminary step; it is my cognitive contribution, and it is the part AI can't yet supply for itself.

The second is that execution to ninety percent is where AI delivers genuine value. Ninety percent completion of a research synthesis that would have taken me weeks, done in hours, is a real capability gain that no honest accounting can dismiss. The productivity gains documented in McKinsey's 2025 State of AI report and PwC's AI Jobs Barometer are real (McKinsey, 2025; PwC, 2025). The organizations I see pulling ahead are those that learned to capture those gains rather than resist them.

The third component is the governed checkpoint, and this is where my thinking has evolved most. Checkpoint-Based Governance started as a decision loop requiring human review before AI output became action. By v5.0, published March 2026, I had worked it into a constitutional framework specifying four properties that distinguish genuine AI Governance from its simulation (Puglisi, 2026l).

Property 1 establishes the primary purpose: CBG is AI Governance, not a quality-check framework or a workflow tool. Its purpose is to supply the governance layer that converts RECCLIN and CAIPR from AI frameworks into governed systems with named human accountability. Every other property builds on that foundation.

Property 2 establishes what I call the unconditional invariant: there is no AI Governance without human authority and accountability. The checkpoint is not where I am optionally present; it is where I am constitutionally required to be present, documented, and accountable. That requirement does not depend on my prior practice or developmental milestones. My authority at the checkpoint is assumed, and the single boundary on that authority is that I cannot direct an AI-assisted outcome that injures a human being.

Property 3 is what I call the injection function: the checkpoint is not where I filter AI output but where my distinctly human intelligence transforms it. My domain

knowledge, contextual judgment, emotional response, creative intuition, and lateral synthesis are things no AI platform produces alone or in combination. The checkpoint is where those capacities enter the work.

Property 4 is the developmental mechanism, and it is the property I am most personally invested in because it is where my governance architecture connects to my measurement work. My longitudinal self-monitoring suggests that practicing CBG through structured AI output review produces cognitive development in whoever is doing the reviewing. Each review cycle builds my capacity to evaluate evidence chains, recognize reasoning gaps, identify suppressed dissent, and perform the synthesis that the ninety-nine percent target requires. CBG v5.0 states this plainly: the checkpoint is not overhead but where augmented intelligence becomes real (Puglisi, 2026l).

Three terms circulate in the current conversation as if they describe the same thing, and they do not. The difference is where authority sits. Ethical AI asks whether something should be done; it is the values layer, principles without enforcement mechanism. Responsible AI applies machine checks to machine output; it can improve AI behavior systematically without requiring that any named human answer personally when it fails. In both, "AI" is the noun, the thing being made ethical or responsible. AI Governance reverses the grammar: "AI" modifies governance, and the human system holds final position. A named human with binding checkpoint authority, backed by personal accountability for outcomes across moral, employment, civil, and criminal channels, is what that grammar reversal requires. That reversal is what I designed CBG to enforce (Puglisi, 2026l).

"Human in the loop" is not AI Governance, and the phrase most organizations use when they want to claim governance without building it is the one I am most careful about. Physical presence at a checkpoint without substantive engagement is not governance; it is the simulation of governance, and CBG exists specifically to prevent that simulation from passing as the real thing.

Harris uses a useful regulatory metaphor in the November 2025 DOAC interview: governance does not require understanding the engine. The FAA does not tell Boeing how to design wings; it requires flight data recorders. The SEC does not tell banks how to invest; it requires audit trails. I developed CBG because I needed that kind of infrastructure for my own work, and I published it so others can test, challenge, and build something better if they can.

What I found when I looked at the Horvath Senate testimony case converged on the same architectural answer from a completely different direction. The policy response to Horvath's binary, which was either remove the technology or surrender, is a three-part standard applicable to any tool deployed in a learning environment: active cognitive demand on the learner, evidence that the deployment method produces the outcome it

claims to produce, and named human accountability with authority to modify or reject any deployment that fails the first two conditions (Puglisi, 2026h).

That three-part standard maps directly onto what my architecture provides: role assignment in the HAIA-RECCLIN structure enforces active cognitive demand, HEQ trajectory measurement provides the evidence that the deployment method is producing what it claims, and CBG provides the named human accountability with binding checkpoint authority. The cognitive science, the governance architecture, and the policy standard all converge on the same structural requirement: a named human with governing authority over the method of use, not just the presence or absence of the tool.

CBG governs what happens at the checkpoint, but structured collaboration also requires a way to compare machine outputs before any single output reaches that checkpoint. The multi-platform dimension of how I actually work has its own name in the stack: HAIA-CAIPR, the human-orchestration protocol for parallel multi-AI execution (Puglisi, 2026k). The principle is straightforward and has been observed consistently in my own practice: independent AI architectures catch different errors.

When I ran eleven-platform content validation for the AI Provider Plurality Congressional Package (a four-document AI governance proposal submitted to the 119th Congress in February 2026) and the GOPEL content review, each platform caught something different or unique that the others missed. When I ran seven-platform adversarial code reviews, the same pattern held. The most instructive documented instance was when eight of nine platforms produced the same incorrect output; the governance process flagged the single dissenter, I triggered verification, and the dissenter was correct. I overrode the eight.

Convergence without dissent is a risk-elevation signal in my methodology, not validation. That is CAIPR in operation: parallel dispatch to independent platforms, individual output collection, human arbitration of divergence, with the human governor holding final authority over every disposition. The adoption ladder I use runs Factics → RECCLIN → CAIPR → GOPEL, with each layer adding governance depth without replacing the layer before it.

The final layer in that stack is GOPEL, the Governance Orchestrator Policy Enforcement Layer, and it addresses a problem CBG cannot solve alone (Puglisi, 2026j). CBG establishes constitutional authority. But authority requires enforcement, and enforcement mechanisms that think can be manipulated. GOPEL is deliberately non-cognitive: it performs seven deterministic operations, dispatch, collect, route, log, pause, hash, and report, with a SHA-256 hash-chained append-only audit trail, and performs zero cognitive evaluation by design. The absence of cognition is the security property: a pipe that can't interpret can't be deceived.

I published a reference implementation, GOPEL v0.6.1, reviewed under adversarial conditions by seven independent AI platforms. Every critical vulnerability identified was fixed and every limitation is documented in the public repository. What GOPEL v0.6.1 establishes is feasibility, not deployment readiness. I coded it to prove the architecture can be built, not to claim the problem is solved (Puglisi, 2026j). That distinction matters to me, and I state it every time I cite it.

One thing I can show rather than claim: this paper was produced using the HAIA governance stack it describes. Multiple AI platforms operating under RECCLIN role assignments drafted, researched, and reviewed sections in parallel. CBG checkpoints governed every substantive decision about what stayed and what changed. A seven-platform CAIPR review of this draft produced the synthesis that drove this version's revisions. The audit trail exists. What I can't yet show is that the process produced a better paper than I would have written alone, and that is precisely what the measurement architecture of Section 8 is designed to eventually answer for work products like this one.

I want to be clear about the epistemic status of all of this. The method these tools implement (governed, structured human-AI collaboration with active cognitive demand, named human accountability, and trajectory measurement) has independent empirical support from the sources in Section 3. The specific tools I developed to operationalize it are working concepts: specified architecture with documented operational evidence, submitted to Congress and published on SSRN, but not yet enacted, peer-reviewed, or production-validated. They represent one implementation path. Other approaches may address the same structural problems through different architecture, and I would welcome them.

Augmented Intelligence is a different frame from the debate the camps are having: they argue about whether AI should be used, and I argue about how the collaboration should be governed and how to know whether the governance is working.

## **The Growth OS: Competition, Not Replacement**

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The Augmented Intelligence architecture I described in Section 4 needs an organizational layer to sustain it at scale. That layer is what I developed in the Growth OS series: the governance and cultural infrastructure that determines whether AI collaboration produces compounding organizational advantage or a sequence of efficiency gains that eventually eliminate the human capacity to govern the collaboration. The job displacement fear that saturates the current conversation follows a logic that sounds grounded but fails to account for those competitive dynamics, and I want to explain what I think is actually happening.

The fear has a real evidentiary base. A 2025 Stanford Digital Economy Lab study found a 16% relative decline in employment for early-career workers aged 22 to 25 in the most AI-exposed occupations, using ADP payroll data covering millions of workers, with software engineering entry-level positions declining by roughly 20% and customer service by roughly 11% (Brynjolfsson et al., 2025). Nearly three in ten companies report having already replaced jobs with AI, and by the end of 2026, 37% expect to have done so (Resume.org, 2025). These are real consequences for real people.

But I argued in the Growth OS series that the relevant comparison is not between humans and AI as substitutes but between organizations that govern human-AI collaboration well and organizations that do not (Puglisi, 2025a). Industries most exposed to AI have nearly quadrupled productivity growth since 2022, and revenue per employee is growing three times faster in the most AI-exposed industries than the least (McKinsey, 2025; PwC, 2025). The organizations generating those gains are not the ones that eliminated their human workforce. They are the ones that restructured the relationship between human judgment and machine execution.

The competitive pressure AI creates does operate as a barrier reduction, and I think this is what the displacement fear correctly identifies. When AI can perform a task that previously required an expensive human specialist, the cost of performing that task falls. I can now compete in domains that were previously accessible only to those with credentials, capital, or years of accumulated experience that I did not have. This creates real pressure on incumbents who built their market position on access to scarce expertise, but it does not create pressure on the humans who learn to direct the collaboration. The barrier reduction is the growth mechanism, not the threat.

Organizations that respond to AI capability by eliminating the human layer are making a bet I think will fail. The voluntary compliance failure pattern I documented in Chapter 2 of *Governing AI* runs directly against it. Companies that announced ethics boards, safety teams, and governance commitments between 2023 and 2024 produced boards meeting quarterly or less, safety teams lacking authority to block deployments, and published principles without implementation protocols enabling verification. The Economic Override Pattern predicts exactly this: when competitive pressure makes safety costly and enforcement mechanisms lack teeth, organizations optimize for the constraints they actually face, not the ones they have promised to observe (Puglisi, 2025b, Chapter 2). The adversarial behavior evidence Anthropic documented across all major platforms confirms the exposure that comes from governance theater rather than governance substance (Anthropic, 2025b).

The Growth OS three pillars I developed, Trust and Transparency, Rhythm and Culture, and Outcome Anchoring, are not soft supplements to technical implementation. They are the architecture that determines whether the technical implementation produces compounding advantage or fragile efficiency gains that collapse under competitive

pressure (Puglisi, 2025a). The forcing question I use with organizations is not whether AI will change the work, because it will. The question is whether you are building the human capacity to govern the collaboration, or outsourcing that governance to the machine and hoping the machine is right.

## **AI as Mirror: The Flaws Are Inherited**

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One of the frameworks I've developed is what I call the Mirror to Humanity, and I think it is the most politically uncomfortable part of my position because it implicates everyone equally.

AI systems inherit the biases of the humans who built them and the data those humans produced. The evidence I've documented is precise: GPT-4's psychological response profile correlates strongly with Western, educated, industrialized, rich, and democratic populations, and weakly or negatively with the rest of humanity (Atari et al., 2023; Henrich et al., 2010). One documented dimension of this is the political composition of academic populations: at elite institutions that produce high volumes of research-quality text used in training, the ideological skew is measurable and has been reported (Harvard Crimson, 2023). The journalism literature carries parallel concerns, though that is a separate evidentiary question. AI systems learn what counts as credible, what requires qualification, and what can be stated without annotation from sources that carry these structural biases. The bias is not random; it is structural and inherited from us.

But the bias risk compounds when combined with human hallucination, the concept I defined in Section 1: stripping methodology from findings and distributing conclusions without the correction mechanism that would let the audience evaluate them. AI hallucination produces a wrong answer the user can verify. Human hallucination produces a wrong interpretation the audience has no mechanism to catch, and the governance response must address both.

The Horvath Senate testimony is the clearest governance-relevant example of this pattern I have found. The same research base cited as support for the biological-determinism claim in the viral statement is also the research base that Horvath's own Chalk and Talk podcast appearance, six days before the Senate hearing, conceded that it supported context-dependent method governance. The binary traveled because the attention economy rewards the simplest, most credentialed version of a claim, and because stripping the methodology from the finding is exactly what makes the finding sharable. The research did not support the conclusion circulating at two million views, and the methodology was the part that was left out (Puglisi, 2026g).

This finding does not support the Never AI position. I have never been willing to refuse a medium because its producers are biased. Books had biased authors and the internet

has biased sources; every medium humanity has used for cognitive collaboration has carried inherited biases, and my response to inherited bias has always been to govern the relationship with the medium rather than abandon the medium.

This finding also does not support the AI Solves Everything position. Deploying AI without governance architecture means deploying the inherited biases of a specific population at scale, without the correction mechanisms that would allow those biases to be identified and addressed. The absence of human oversight is not neutrality; it is the distribution of the biases embedded in the training data, including the methodological biases that produce human hallucination when distributed without accountability.

My governance response is the same one the collaboration arc has always required: structured accountability for the outputs produced by the collaboration between human and machine. When I review AI output at a governed checkpoint and take responsibility for the result, my review applies judgment that can catch the inherited biases the AI cannot detect in itself. The checkpoint is the correction mechanism, and its absence is not efficiency. It is the substitution of machine bias for human oversight at the accountability layer.

## **Breaking Socioeconomic Barriers: Why This Matters Beyond Business**

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One dimension of the Augmented Intelligence argument I find most compelling is one I rarely hear discussed.

The collaboration arc has always produced asymmetries of access. Books were available to the literate and, for most of history, to those with sufficient wealth to own them. The internet reduced the cost of access to information but did not eliminate the skills required to evaluate and use it. Social media gave everyone a publishing platform but concentrated the amplification mechanisms in the hands of those with existing networks and resources.

AI changes the access equation in a way that I think is genuinely different from prior steps in the arc. I am an independent practitioner who did not come from an academic institution with a large research infrastructure. The work I have been able to do, building open-source governance frameworks, submitting Congressional packages, publishing on SSRN, engaging thought leaders on their own published terms, is work that AI has made possible for me at a speed and depth that simply was not available before. I am one data point, but the direction of the pressure is clear: AI reduces the cost of accessing cognitive assistance in domains where the cost was previously prohibitive for large portions of humanity, a direction of pressure the World Economic Forum's 2025 Future of Jobs Report identifies as one of the most consequential shifts in how capability will be distributed globally (World Economic Forum, 2025).

A first-generation college student in a country without strong research infrastructure can now access synthesis and analysis capabilities that were previously available only to researchers at well-resourced institutions. A small business owner without the capital to hire a legal specialist can use AI to understand contract language that would otherwise require expensive professional consultation. A practitioner in a developing economy can access the accumulated knowledge of the global professional literature in a fraction of the time and cost that access previously required.

I am not claiming the access is equal or that the remaining barriers are trivial. The concerns Hao raises about who benefits from AI development are legitimate, and the gap between the promise of access and the reality of access is real (Puglisi, 2026a).

But the direction of the pressure is unambiguous, and the governance question for me is not whether to allow that access but how to structure it so that the reduction of barriers doesn't simultaneously create new forms of dependence or exploitation.

The socioeconomic case for governed Augmented Intelligence is ultimately an argument about what the collaboration arc has always delivered at its best: the extension of cognitive capability beyond the boundaries previously set by economic circumstance. Books did this at the scale of literacy, and the internet did this at the scale of connectivity. AI can do this at the scale of expertise, if the governance architecture prevents the concentration of that capability in the hands of the few who control the training infrastructure.

## **What I Built to Measure It: HEQ and AIS**

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A collaboration architecture that can't be measured can't be improved, and one that can't show growth in me over time doesn't actually produce the outcome it promises. What I developed to address this measurement gap also went through an evolution that is itself part of the Augmented Intelligence story.

The instrument that became the Human Enhancement Quotient started as the Factics Intelligence Dashboard, published August 2025, designed to measure in-session applied intelligence, how intelligence performs during actual work, rather than abstract capability. The foundational principle I started with was this: IQ was built to measure intelligence in isolation, while what I needed was an instrument that measured intelligence in context (Puglisi, 2026d). When I administered the instrument across five architecturally distinct AI systems simultaneously, I got a cross-platform consistency coefficient of  $ICC = 0.96$ , which told me the approach was at least feasible.

The theoretical lineage the instrument draws from is not mine. Licklider (1960) and Engelbart (1962) established that augmentation of the human-plus-tools system, not substitution of one by the other, is the productive frame. Kasparov (2017) provided the

empirical foundation through Advanced Chess: the human contributor's quality of judgment, not the machine's computational power, determined team success. Noy and Zhang (2023) extended this into the generative AI era, showing that productivity gains from AI collaboration are real but do not answer whether capability remains when AI is removed, which is precisely what HEQ is designed to measure.

Dellermann, Ebel, Sollner, and Leimeister (2019) formalized hybrid intelligence as a discipline. Dweck (2006) established that intelligence grows through challenge, creating the theoretical necessity of trajectory measurement rather than snapshot assessment. I assembled these into an instrument because none of them individually produced what I needed: a single measurement that tracks whether the human is getting stronger or weaker through the collaboration over time.

The critical shift came during that testing phase. I documented something in October 2025 that changed the entire direction of the work: the measurement itself was producing growth. The subjects were not only performing better when they used AI; their rubric performance improved under repeated structured assessment. That discovery shifted my design from measuring performance to measuring enhancement (Puglisi, 2026d). The question I started asking was no longer "how well does this person collaborate with AI right now" but "is this person getting cognitively stronger or weaker as a result of the collaboration over time."

The four dimensions I settled on were not assumed theoretically. I started with six and consolidated them through iterative empirical testing as I found which patterns tracked the same underlying process. The four that survived are Cognitive Agility Speed (how quickly and clearly I process and connect ideas under AI-augmented working memory load), Ethical Alignment Index (how consistently my thinking reflects fairness, responsibility, and transparency under uncertainty), Collaborative Intelligence Quotient (how effectively I integrate diverse perspectives within AI-augmented collaboration), and Adaptive Growth Rate (the trajectory of capability improvement through sustained AI collaboration) (Puglisi, 2026d).

AIS is the composite output, expressed as  $(CAS + EAI + CIQ + AGR) / 4$ . I chose the arithmetic mean because a weighted composite requires empirically derived factor loadings from validation studies I haven't completed yet, and equal weighting is the epistemically honest choice against which future empirically derived weightings can be compared.

One finding has been consistent enough that I want to name it: Collaborative Intelligence Quotient is the lowest-scoring dimension across everyone I have administered the instrument to. Cross-user testing of ten people identified CIQ as lowest in all ten cases. The five original platforms in my 2025 baseline produced the same pattern.

This aligns with independent research on human over-reliance on AI, specifically Lane et al. (2025) on the dangers of deferring to AI and Vaccaro, Almaatouq, and Malone (2024) on conditions under which human-AI combinations underperform. Their meta-analysis of 106 studies found that on average, human-AI systems performed worse than the best of either alone when the interaction was unstructured. I read this as pointing to the exact capacity the collaboration architecture must develop: appropriate reliance, the ability to calibrate when to trust the AI's output and when to push back.

The CBG developmental mechanism completes the loop. My own ten-month longitudinal measurement (a single-practitioner self-monitoring record,  $n=1$ , not a controlled study) found that CIQ showed the largest dimensional gain of any measured dimension, rising from 88.4 to 93.4, consistent with ten months of structured checkpoint governance practice designed to build that calibration capacity (Puglisi, 2026f). I state  $n=1$  directly because burying that qualification in a disclaimer section while presenting the numbers in the body would be the kind of methodological stripping I criticize elsewhere. The directional pattern is worth noting; the evidential weight is exactly what single-practitioner self-monitoring can carry, no more.

What CBG builds in the human practitioner, HEQ measures and AIS expresses. The governance architecture and the measurement instrument are not separate tools; they are a reinforcing system where the checkpoint practice is the training mechanism and the AIS trajectory is the evidence that the training is working.

I need to be honest about the limitations here. HEQ requires formal psychometric validation that I have not completed, and no independent research group has yet replicated my administration or validated the instrument against external criteria. Anyone who adopts it now becomes a validation partner whose data will strengthen or challenge my measurement case. I state this openly because the Empire of Evidence paper I published identified governance risk when companies hide their methodological gaps, and I will not do the same thing (Puglisi, 2026a).

## **What I Believe and Why It Matters**

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The AI race Harris documented is real. The extraction architecture Hao documented is real. The cognitive decline concern entering federal policy is producing governance from claims the evidence cannot yet support. All three deserve the infrastructure their strongest findings require, and none of the three has produced it. Harris's warning reaches hundreds of millions but does not yet have architecture for policymakers to act on. Hao's evidence is primary-source quality but implies a prescription the collaboration arc's full history does not support. The cognitive decline narrative has stripped methodology from a contested research base and is traveling the binary conclusion as settled fact (Puglisi, 2026g; Puglisi, 2026h; Puglisi, 2026i).

The Bastani finding states what I believe the evidence shows: the same AI tool used without structure reduced exam performance by 17% when removed, while the same AI tool used with structured scaffolding nearly eliminated the negative effect (Bastani et al., 2025). The variable is not the technology. The variable is whether the human governing the interaction has been given the architecture to make the collaboration developmental rather than substitutional. That is the operative question, and it is the question my work is built to answer.

I am not going to stop using AI because the companies that built it are imperfect. I did not stop reading books because printing houses concentrated the distribution of knowledge. The medium evolves, the collaboration need persists, and the governance architecture either develops to match the medium or the errors compound until an institutional failure forces a correction.

What I've developed is my answer to the governance question: I provide the structure, AI executes to ninety percent, and my governed checkpoint closes to ninety-nine (again, a conceptual target, not a measured completion score), with me accountable for the result and the four accountability channels intact: moral, employment, civil, and criminal. My measurement framework tracks whether I am growing through the collaboration or degrading. My Growth OS captures the organizational dynamic that turns the collaboration into competitive advantage rather than efficiency gain that eliminates its own workforce.

None of this is proven. All of it is published so others can test, challenge, improve, or replace it. The principle I am building from is not mine alone: sixty years of augmentation research, Licklider to Engelbart to Dellermann, has been pointing at the same thing. I am one practitioner who took those principles and created infrastructure around them because I needed them for my own work, and because I could not find anyone else who had built what I needed.

Three pressures are pushing the public conversation toward a binary that cannot produce a useful answer. Harris asks for clarity and political will, Hao asks for accountability and structural reform, and the cognitive decline narrative asks for removal. My position is different from all three. I ask who is governing the method, under what architecture, with what audit trail, and how we know whether the collaboration is making us better or worse.

If you are a policymaker, I am asking you to look at CBG as the accountability layer that AI regulations need to specify: not principles, not ethics boards, but named human authority with binding checkpoint authority and the four accountability channels that make individual behavior respond to consequences.

If you are a researcher, I am asking you to test HEQ, replicate its administration, challenge its dimensions, and tell me where the measurement fails, because the only way the instrument gets better is if someone outside my own practice runs it.

If you are a practitioner, I am asking you to try the HAIA-RECCLIN structure on your next AI-assisted project, run more than one platform, preserve the dissent, and track whether your governing capacity improves over the following quarter. The only way to know whether any of this works is to use it, measure it, and iterate.

That is the only question worth arguing about at scale, and it is the question I intend to keep answering.

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